

Ilaria Lunghini

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Education

Università degli Studi di Verona

Graduated July 2026

Bachelor's Degree in Ingegneria dei Sistemi Medicali per la Persona (106/110)

Verona, Italy

- **Thesis:** Development of a pipeline for connectivity analysis during EEG microstates in healthy controls and patients with multiple sclerosis.
- **Relevant Coursework:** Analisi e modellazione dei sistemi biologici; Analisi del movimento; Chirurgia robotica e neuroradiologia.

University of Padua

MSc student, first year

Master's Degree in Biomedical Engineering for Neuroscience

Padua, Italy

Experience

BraiNavLab

Dec 2025 – Jun 2026

Biomedical Engineer, Thesis Internship

Verona, Italy

- Developed a MATLAB pipeline using EEGLAB and FieldTrip to analyze EEG microstate temporal dynamics and functional connectivity in healthy controls and individuals with multiple sclerosis.
- Implemented clustering, AEC orthogonalization, Wilcoxon rank-sum tests, and graph-theoretical analysis focused on degree, closeness, betweenness, eigenvector centrality, and the clustering coefficient.
- Compared pipeline outputs with published literature and state-of-the-art approaches to contextualize exploratory findings; observed trends aligned with previous EEG connectivity research.
- Produced visualizations, including BrainNet Viewer network visualizations, a technical report, and a website documenting the pipeline workflow and main analysis steps.

Projects

EEG Source Connectivity Analysis | MATLAB, FieldTrip, eLORETA

- Collaborated on a source-level EEG analysis workflow, applying a three-layer BEM head model, electrode alignment, lead-field computation, and eLORETA source reconstruction.
- Applied atlas-based parcellation and delta-band partial directed coherence (PDC) analysis to examine directional interactions across 100 brain regions.
- Generated connectivity matrices and BrainNet Viewer network visualizations to examine weighted node strength.

Gait Kinematics with Joint Coordinate Systems | MATLAB, SVD-based Registration

- Co-developed a MATLAB workflow to reconstruct tibia, calcaneus, metatarsus, and anatomical-foot motion from gait marker trajectories.
- Implemented segment coordinate systems, SVD-based rigid-body registration, and joint-coordinate-system decomposition to calculate three-dimensional joint angles.
- Normalized joint-angle trajectories across gait cycles and generated plots of plantar/dorsiflexion, internal/external rotation, inversion/eversion, and sagittal angular velocity.

Task-Based fMRI GLM Analysis | MATLAB, SPM12

- Co-developed a voxelwise general linear model (GLM) workflow for task-based fMRI data comparing motor execution and motor imagery.
- Implemented canonical-HRF regressor construction, DCT-based temporal high-pass filtering, and ordinary least-squares model fitting within a brain mask.
- Generated condition-specific t-statistic maps and thresholded activation overlays on mean fMRI anatomy.

Technical Skills

Programming: MATLAB, Python

Neuroengineering & Simulation: MNE-Python, EEGLAB, FieldTrip, MICROSTATELAB, SPM12, BrainNet Viewer, NEST Simulator

Scientific Computing: NumPy, Pandas, Matplotlib, HDF5/h5py, SONATA

Methods: EEG microstates, source reconstruction, functional and directed connectivity analysis, graph-theoretical analysis, fMRI GLM, gait kinematics, spiking neural-network simulation

Robotics & Simulation: ROS 2, Gazebo, CoppeliaSim, CAD

Development: Git, GitHub

Certification: Pattern Strategici di Problem Solving, Università degli Studi di Verona (2026)